

An efficient feature selection method for arabic and english speech emotion recognition using Grey Wolf Optimizer

Ismail Shahin^c, Osama Ahmad Alomari^a, Ali Bou Nassif^d, Imad Afyouni^b, Ibrahim Abaker Hashem^b, Ashraf Elnagar^{b,*}

^a MLALP Research Group, University of Sharjah, Sharjah P.O. Box 27272, United Arab Emirates

^b Computer Science Department, University of Sharjah, Sharjah P.O. Box 27272, United Arab Emirates

^c Electrical Engineering Department, University of Sharjah, Sharjah P.O. Box 27272, United Arab Emirates

^d Computer Engineering Department, University of Sharjah, Sharjah P.O. Box 27272, United Arab Emirates

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ABSTRACT

Nowadays, analyzing and interpreting emotions through human speech communication have drawn a great attention in the field of human-computer interaction. Therefore, many speech recognition systems have been suggested to recognize the emotional states of the speaker utilizing the speech recordings of their spoken utterances. Feature extraction is an important step in building an emotion recognition system in which it is used to extract emotional features from speech data. However, not all extracted features are relevant to classify the emotion states of the speaker. The existence of irrelevant and redundant features generates unmeaningful patterns that lead to inaccurate and undesirable emotion classification performance. Therefore, this study proposes an intelligent feature selection method based on a novel bio-inspired optimization algorithm that mimics the hunting mechanism of wolves in the nature, called Grey Wolf Optimizer (GWO) and K-nearest neighbor (KNN) classifier, to find the most relevant subset of features to enhance the classification performance of an emotion recognition systems. The proposed method is called GWO-KNN. Emotion classification is performed on three distinct databases including Arabic Emirati-accented speech database, Ryerson Audio-Visual Database of Emotional Speech and Song dataset (RAVDESS), and Surrey Audio-Visual Expressed Emotion dataset (SAVEE). A combined or single feature extraction method is applied to extract the features from each dataset. The proposed method provides better classification performance for speech emotion recognition system compared to classical methods such as bat algorithm (BAT), cuckoo search (CS), White Shark Optimizer (WSH), and arithmetic optimization algorithm (AOA). Our proposed method also surpasses several state-of-the-art recent approaches that use the same datasets.

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1. Introduction

Speech signal has emerged as a predominant tool for human communication. Thus, speech emotion recognition has recently presented a significant challenge in the field of human-computer interaction (HCI) due to its ambiguous features [1]. The main goal is to analyze audio features to determine a person's current emotional state [2]. Several applications of speech emotion recognition have been recently introduced, including speech processing, audio surveillance, detection of lies, computer games, etc [3]. However, extraction of speech features problem remains significantly difficult for emotion recognition in speech signals [4] proposed a

new feature selection method based on the combination of Hilbert-Huang Transform (HHT) based visual features, HHT-based audio features, and cross-correlation features. The proposed method enhanced the performance of video effective recognition. Moreover, [5] proposed modulation spectral features for speech recognition that automatically recognizes human affective information. Their proposed method improved the recognition for classifying seven emotion categories.

Conventionally, emotion recognition system is divided into three main components: feature extraction, feature selection, and emotion recognizer [6]. However, researchers stated that the performance of emotion recognition system is affected by two factors that are: how to find the most relevant emotion features and how to build an accurate speech emotion recognition model [7]. Identifying informative, discriminative, and meaningful features is an

* Corresponding author.

E-mail address: ashraf@sharjah.ac.ae (A. Elnagar).

essential step in classifying emotional states [8]. This process is called feature selection. The main target of feature selection is to select a subset of relevant emotional features from a set of extracted features that can boost the classification accuracy while discarding the irrelevant and noisy emotion features. In the literature, many metaheuristic and exhaustive search algorithms have been applied for solving feature selection in emotion recognition and other fields [9–11]. In these studies, the proper selection of the most relevant subset of features from the raw acoustic features has proven to be effective in dimension reduction of the training data and the enhancement of emotion classification. Nevertheless, the problem of premature convergence that was faced by the optimization feature selection methods poses a challenge toward having the best speech recognition system [12,10,13].

In recent years, Grey Wolf optimizer (GWO) stood out as a promising metaheuristic intelligence swarm technique which was introduced by [14] to imitate the leadership hierarchy of gray wolf packs and the hunting mechanism. The optimization searching procedure of GWO is guided by three features inspired from the behaviour of Gray wolf packs, which are the way of: seeking the prey (exploratory search operator), encircling the prey, and the attacking (exploitative search operator). The harmony among these operators plays a crucial role in avoiding any potential premature convergence. Compared to other swarm intelligent optimization algorithm, GWO has the following merits: it has no parameters to tune, easy to implement and adapted for optimization problems, flexibility, and scalability. Therefore, in recent years, GWO has been widely applied as a feature selection technique in different domains such as intrusion detection [15], big data analytics [16], image classification [17], facial emotion recognition [18], EEG signal classification bioinformatics [19], and bioinformatics [20]. Motivated by the attractive features and impressive performance of GWO, this paper proposes a new feature selection method based on GWO for emotion recognition system. The main contributions of this work are summarized as follows:

- An improved speech emotion recognition system is proposed using an adapted GWO as the feature selection technique and KNN algorithm for the classification task.
- The performance of the proposed method is tested on a local real constructed dataset and two benchmarked-datasets. Namely, Arabic Emirati-accent dataset [21], Ryerson Audio-Visual Database (RAVDESS) [22], and SAVEE database [23].
- The convergence rate of GWO is evaluated against recent and well-known optimization algorithms.
- Extensive experiments have been conducted to show the outstanding performance when compared with other competitive algorithms on all experimented datasets.

The remainder of the paper is structured as follows: Section 3 discusses the related work. Section 3 presents the fundamentals of Grey Wolf Optimizer. Section 4 provides material and methods. Section 5 provides data preparation. Section 6 presents results and discussion. Section 7 shows concluding remarks and future work.

Human speech generally combines the emotional states of an individual along with the syntax and semantic content of expressed sentences. Speech emotion recognition (SER) consists of classifying speaker's emotions by analyzing their speech recordings, thus inferring and predicting their physical and psychological state. In this section, we overview the main related work on speech emotion recognition, their feature extraction and/or selection methods, their classification techniques, the datasets used, and their performance against existing benchmarks.

Vocal Emotion Recognition has been studied with a focus on classical machine learning algorithms, such Support Vector Machine (SVM) and multiclass SVM, Random Forests, and

Adaboost [24]. Feature selection is a key component in speech emotion recognition [25]. The authors in [1] proposed a hybrid meta-heuristic feature selection method using Golden Ratio Optimization (GRO) and Equilibrium Optimization (EO) algorithms, that are fed into an XGBoost classifier. Linear Predictive Coding (LPC) [26] and Linear Prediction Cepstral Coefficient (LPCC) [27] features were selected as input for the optimization algorithm. Their model claimed to perform better than other machine learning (ML) and deep learning methods.

A Whale-Imperialist Optimization algorithm (Whale-IpCA) based on Multiple Support Vector Neural Network (Multi-SVNN) classifier was introduced in [28] as a new emotion recognition scheme for identifying emotions in speech signals. The idea is to use a spectral feature set as an input signal to identify emotions. Yildirim et al. [11] proposed a feature selection method for meta-heuristic search algorithms that modified the initial population generation stage. The authors used two popular metaheuristic search algorithms named Cuckoo Search and NSGA-II to evaluate speech emotion recognition on Interactive Emotional Dyadic Motion Capture and Berlin emotional speech database. The presented feature selection algorithms significantly reduced the number of features while still being effective for emotion classification from speech, according to the results.

Kerkeni et al. [29] introduced a comprising speech emotion recognition system based on various machine learning algorithms such as Recurrent neural networks (RNNs), Support vector machines (SVMs), and Multivariate linear regression (MLR) and feature extraction methods for affective state and the modalities of emotion expression. The experiment was performed on Berlin database of emotional speech which showed an accuracy of 83% on speaker normalization (SN) and a feature selection; however, they showed an accuracy of 94% when performed on the Spanish database using an RNN. Moreover, Issa et al. [3] developed an architecture to extract various features such as Tonnetz representation, chromatogram, Mel-frequency cepstral coefficients, etc from speech signals. The proposed architecture adopted Convolutional Neural Network (CNN) to detect emotions using samples from the RAVDESS database. SVM was also been used by [30] to extract features such as pitch, linear prediction cepstrum coefficients, Mel Frequency cepstrum coefficients (MFCCs), and energy from the Berlin emotional database to automatically recognize speech emotions using different emotional states. The authors adopted an intelligent-based technique to accurately classify these features into distinct emotional states.

An enhanced cat swarm optimization (ECSO) algorithm has been proposed in [31] for optimal feature selection, with the aim of reducing redundant features and high computational cost. The basic CSO algorithm mimics the cat behavior, where position, velocity, fitness value, and seeking or tracing mode, are reported to solve the optimization problem. In addition, to improve the population diversity and search ability of the CSO algorithm, the authors proposed to use Opposition Based Learning (OBL) method. An SVNN classifier was implemented on top of the ECSO method, and the results showed an excellent recognition rate.

The use of ensemble models such as bagging and boosting with SVMs for emotion recognition from speech was investigated in [32]. Bagging, also known as bootstrap aggregation, consists of training various data samples on different base classifiers, and then infer the best estimator by using majority voting rule. A bagged ensemble of 20 SVM estimators with a Gaussian kernel were trained and experimented in the model described in [32]. They used various types of features, such as (MFCCs), spectral centroids and MFCC derivatives. The results were presented with three diverse emotional speech corpora: Berlin EmoDB [33], RAVDESS [22], and IITKGP-SEHSC [34]. Their ensemble model showed an increase of 5% of accuracy over the simple SVM classifier.

Moreover, it outperformed AdaBoost ensemble's performance for all the three datasets and all emotion classes.

A focus on dimensionality reduction of speech features using weighted binary cuckoo search was proposed in [9]. The authors argued that existing intelligent optimization algorithms are optimized for discrete problems rather than continuous speech signals. Therefore, their weighted algorithm considered two stages, where an ensemble learning model based on the random forest algorithm was first used to infer the importance of each feature, and applying the existing binary cuckoo [35], by combining it with multiple classifiers. Their results showed that a combination of Logistic Regression (LR) and weighted binary cuckoo search (WBCS) achieved the best performance. The performance of Cuckoo Search and nondominated sorting genetic algorithm-II (NSGA-II) metaheuristics search algorithms for feature selection was investigated in [11]. Their aim was to generate a set of initial populations of features based on their relevance weights, thus reducing significantly the number of features, while being as effective as other approaches.

A clustering-based genetic algorithm was presented in [36] for feature selection optimization. Clustering in this method was employed at the fitness evaluation level to determine any outliers that could be excluded in the next generation phase. Their method achieved an accuracy of 89.6%. In a different approach, a modified quantum-behaved particle swarm optimization algorithm (QPSO) was proposed in [37] to tackle dimensionality reduction and classifier parameter selection problems. A Gaussian Mixture Model (GMM) was used for classifying emotions. Their methods also relied on the strong correlation between glottal waveform features and the speaking style of an individual, which resulted in a more accurate recognition performance. However, QPSO variants generally suffer from high complexity and slow convergence speed, so the training curve was very high, but the authors claimed it can be used for real-time applications.

Deep learning has been recently employed for MFCC's spectrogram feature selection [38]. Particularly, CNNs and Capsule neural networks (CapsNet) have been investigated for performing recognition tasks [39]. CNN could be developed to learn low-level features such as pitch and energy, while deeper layers can detect more complicated features. However, due to the lack of spatial information in spectrograms, CNNs showed some limitations in this regard [40]. CapsNet, proposed by Sabour et al. [40] have gained popularity as they overcome the shortcoming of CNNs, by embedding orientation and other precise instantiation parameters in their capsules addition to a dynamic routing algorithm to pair activity vectors between lower and upper layers of the capsules. The performance of using CapsNets in recognizing emotional speech has been recently studied in [38]. The results were evaluated using three speech databases (Arabic and English), including, the Emirati Speech Database, SUSAS Dataset, and RAVDESS. Their experimental analysis demonstrated the superiority of the CapsNet model over baseline systems. However, due to the complexity and slow training of CapsNet, the authors in [38] introduced a dual-channel LSTM compressed-CapsNet to better adapt to speech environments, and was tested with different classical classifiers. The proposed model claimed to attain higher accuracy compared to traditional CapsNet. More applications can be found in [41–43].

Based on the previous related work, the use of optimization algorithms as feature selection techniques in speech-emotion recognition systems has produced promising results in terms of boosting classification accuracy and reducing the number of selected features. However, the optimization algorithms suffer from the problem known as “no free lunch” theorem, i.e., no particular optimization algorithm works best across all optimization problems [44]. Hence, this means that existing optimization feature selection methods experienced a downgrade in speech emotion recognition performance. Therefore, there is still a window

to investigate other optimization algorithms. The merits and the strong searching capabilities of the Grey Wolf optimizer (as discussed in Section 1) motivate us to propose it as a feature selection technique in the speech emotion recognition domain.

2. Related work

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3. Research background

This section provides detation about Grey Wolf Optimizer and K-Nearest Neighbors.

3.1. Grey Wolf Optimizer

Mirjalili et al. [14] mathematically designed GWO to replicate the hunting process and social hierarchy of grey wolf packs.

3.1.1. Social hierarchy

Grey wolves go in accordance with a cohesive cultural dominance order which involves the alpha, beta, delta, and omega categories. At the top of the hierarchy is the alpha, which is considered a dominant authority. It is however the smartest when it comes to management of the pack. The alpha wolves are assisted in ruling the pack by the beta wolves, which are as well decision makers at the second social hierarchy level. The lowest of the three mentioned categories is the Omega, which houses most of the pack wolves. The Delta wolves are not part of any of the aforementioned levels. They appear next after the beta wolves in hierarchy and are leaders of the member of the omega level. The GWO solutions are classified according to the grey wolf social order. The alpha (α) wolf is regarded as the fittest of all, then the beta (β), delta (δ) and omega (ω) wolves. Fig. 1 illustrates the social hierarchy of the wolves in a pack.

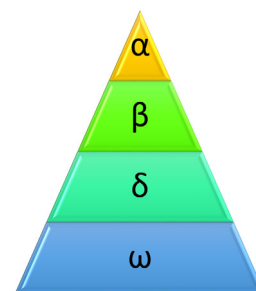


Fig. 1. GWO social hierarchy.

3.1.2. Encircling prey

The hunting behaviour of the wolves in a pack is a smart method which is formulated. This behaviour is of three main stages: chasing, encircling, and attacking. A wolf has the ability to change its location by surrounding the prey to an area within the position of the prey. This is the word used to describe the grey wolves' encircling behaviour, which can be defined as:

$$\mu = |v \times X'(i) - X(i)|, \quad (1)$$

$$X(i+1) = X'(i) - \eta \times \mu, \quad (2)$$

where η and v denote two controllers, X' stands for the location of the prey, X is the wolf location, and i signifies the present iteration. The two controllers (i.e., η and v) are calculated using:

$$\eta = 2 \times \iota \times r_1 - \iota, \quad (3)$$

$$v = 2 \times r_2, \quad (4)$$

where ι is linearly declines from 2 to 0 over the iterations, and r_1 and r_2 are derived at random in $[0, 1]$ interval. The range of value of η falls within the interval $[-2\iota, 2\iota]$ that is dependent on the values of ι .

3.1.3. Search for prey (exploration)

The location of the alpha, beta, and delta wolves has an effect on the searching behaviour adopted by the remaining wolves in finding their prey. In search for prey, the wolves separated from each other and then come together so as they might encircle and attack a prey. The wolves' divergence is represented by η with values that change at random to values that are higher than 1 or lower than -1 to push the movement of search agent in away from the prey location. This procedure is aimed at ensuring a thorough exploration and enhancing the extensibility of the GWO to achieve a global search as well as the most desired solution.

The v in GWO is in charge of the exploration phase. It comprises of the random values that fall within 0 and 2 which enable the prey to emphasize or deemphasize at random, where v exceeds or is less than 1 accordingly.

As compared to η , the v is not known to undergo a linear diminishing. The GWO provides random values whenever there is a need to emphasize exploitation and exploration by the use of v including during the final iteration. The attributes is of great importance especially in instances whereby the search agents could not move out of a local optima, particularly during the last iterations.

3.1.4. Attacking prey (exploitation)

A mathematical model of surrounding the prey is given through a linear deduction of the ι value. η 's fluctuation range is similarly decreased by the ι parameter. The location of the search agent can be updated to a site close to the location of the prey in a situation where η comprises of random values that fall within 1 and -1.

3.1.5. Hunting

Due to the dominant nature of the alpha and the leadership position it assumes in the pack, it controls the hunting activities that take place in the pack. Nonetheless, the beta and delta wolves sometimes take part in the hunting. Owing to that, the 3 superior solutions are represented by alpha, beta, and delta. Other search agents in the pack are thereby expected to make a shift in their location to match the assumed locations of the three superior wolves (i.e., alpha, beta, and delta). The hunting mechanism is formulated as shown below:

$$\mu_\alpha = |v_1 \times X'_\alpha - X| \quad (5)$$

$$\mu_\beta = |v_2 \times X'_\beta - X| \quad (6)$$

$$\mu_\delta = |v_3 \times X'_\delta - X| \quad (7)$$

$$X_1 = X'_\alpha - \eta_1 \times \mu_\alpha \quad (8)$$

$$X_2 = X'_\beta - \eta_2 \times \mu_\beta \quad (9)$$

$$X_3 = X'_\delta - \eta_3 \times \mu_\delta \quad (10)$$

$$X(i+1) = \frac{X_1 + X_2 + X_3}{3} \quad (11)$$

Alpha, beta, and delta wolves may assess the position of the prey so as to update the locations of the other search agents to be fit for hunting operations.

3.2. K-nearest neighbors

kNN is a non-parametric algorithm implemented to perform classification and regression tasks [45,46]. In classification, kNN relies on the information gained from k nearest neighbors to predict new samples, so that the assigned class is the one that has high frequency among other classes. Fig. 2 presents a simple example of kNN procedure with several k values. As can be seen, the prediction achieved with $k = 3$ would be Y, with $k = 5$ would be X and with $k = 11$ would be X.

The most important merit of this algorithm is that it does not construct a model for performing the prediction task. Usually, there is no action until the unlabeled data patterns are discovered, thus the denomination of lazy approach [47]. Once the sample to be classified is provided, the information gained by its k nearest neighbors is utilized as elaborated above [48].

4. Material and methods

4.1. Emotional databases

A fundamental and crucial step in tackling any research problem is to collect a suitable dataset. In our case, we utilized two publicly available benchmark datasets, specifically SAVEE and RAVDESS, as well as a local speech dataset called Emirati-accent dataset.

4.1.1. Emirati speech database

The Arabic Emirati speech database [49] was articulated by 9 male and 22 female local Emirati speakers ranging in age from 14 to 55 years. Every speaker uttered eight public Emirati sentences that are widely used in the "United Arab Emirates" culture. The eight sentences were repeated nine times by each narrator in

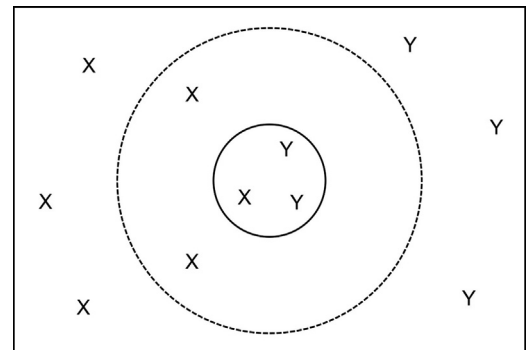


Fig. 2. k-Nearest neighbors algorithm in a bi-dimensional space.

Table 1
Samples of selected features.

No.	English Version	Emirati Accent
1.	I'm leaving now, may God keep you safe.	فداعة الرحمن بترخص عنكم الحينه.
2.	The one whose hand is in the water is not the same as whose hand is in the fire.	اللي ايده في الماي مب نفس اللي ايده في الضو.
3.	Where do you want to go today?	وين تبون تسيرون اليوم؟
4.	The weather is nice, let's sit outdoor.	قوموا نيلس في الحوي، الجو غاوي برع.
5.	What's in the pot, the spoon gets it out.	اللي في الجدر يطلعه الملاس.
6.	Welcome millions, and they are not enough.	مرحبا ملايين ولا يسدن.
7.	Get ready, I will pick you up tomorrow.	زهب عمرك بخطف عليك باجر.
8.	Who does not know the value of the falcon, will grill it like a chicken.	اللي ما يعرف الصقر يشويه.

each of the “neutral, happy, sad, disgust, angry, and fear emotions” over a period of 2–5 s. Speakers lacked the necessary competence to avoid falsifying language. Table 1 lists the eight sentences (“the right column contains the sentences in Emirati emphasis, while the left column contains the English version”). This data was collected within two distinct as well as separate phases: “instructional training session and testing session”.

4.1.2. RAVDESS corpus

The RAVDESS [22] is an endorsed database consisting of both speech and song audio. In the multimodal and gender-balanced corpus, 24 (12 male and 12 female professional actors) participated in vocalizing corresponding etymological statements captured under a neutral North American intonation. Both captured speech and songs contain the expressions: sad, angry, calm, happy, fearful, disgust, and surprise. There are two stages of emotional intensity in each expression which are encompassed, with a disinterested expression in between. Each 7,356 recordings was assessed ten times for emotional validity, intensity, and authenticity. There were 247 untrained research volunteers from North America supplied ratings. A further 72 individuals supplied test–retest data. For this work, only 1,440 speech audio files have been used while 3,036 song files have been ignored (1,012 audios are in the song files and 4,904 are audio-visual files).

4.1.3. SAVEE database

SAVEE [23] is a database of 480 English utterances captured with the aid of four native English male speakers at the “University of Surrey” ranging in age from 27 to 31 years, portraying seven distinct sentiments: contempt, anger, happiness, sorrow, fear, neutral, and surprise. The text content is comprised of 15 TIMIT phrases for each emotion: 3 ordinary, 2 emotion-definite, and 10 generalized lines that are phonetically balanced and distinct for each emotion. To make 30 neutral sentences, the 3 common and $2 \times 6 = 12$ emotion-specific sentences are classified as neutral. This results in 120 utterances per speaker.

4.2. Feature extraction

Speech signals contain a plethora of characteristics known as speech features. Identifying these characteristics is a critical stage in the “speech signal processing” model that affects the overall performance of the proposed system accuracy [50]. In the proposed framework, “Mel Frequency Cepstral Coefficients (MFCC), Zero Crossing Rate (ZCR), Chroma, Mel spectrogram, and Root Mean Square (RMS)” are the speech features that have been extracted in this work.

4.2.1. Mel-frequency cepstral coefficients

One of the main techniques for feature extraction is to utilize MFCCs [50–52]. The most prominent disadvantage of employing MFCCs is its susceptibility to distortion due to its spectral structure dependency. Although speech contains aperiodic content that employs data in the frequency of signals of the speech might be utilized to conquer this challenge [53]. The MFCCs consist of a pair of filters that are linearly set at frequencies less than 1,000 kHz and logarithmically at frequencies more than 1,000 Hz [54,55]. Fig. 3 depicts the MFCCs computation. The MFCCs are computed using the below formula [56]:

$$c(n) = \sum_{m=1}^M [\log y(m)] \cos\left[\frac{\pi n}{M} \left(m - \frac{1}{2}\right)\right] \quad (12)$$

where $y(m)$ denotes the output of the M -channel filter bank and n denotes the index of the cepstral coefficient.

4.2.2. Zero crossing rate

The total number of occurrences in which the sound signal oscillates or switches from positive to negative and vice versa is measured by the Zero-Crossing Rate (ZCR). Specifically, ZCR measures the number of instances that occur during a certain time span or frame, during which the amplitude of the audio signal gradually decreases to zero at some point during the transmission. The dominant frequency part of the signal can be appraised by the ZCR [57–59]. The ZCR is acquired in this manner [60]:

1. Let $x_i(n)=0,1 \dots, N-1$ depicts the i th frame samples.
2. Compute the ZCR for the i th frame as:

$$Z(i) = \frac{1}{2N} \sum_{n=0}^{N-1} |\text{sgn}[x_i(n)] - [x_i(n-1)]| \quad (13)$$

where,

$$\text{sgn}[x_i(n)] = \begin{cases} 1 & x_i(n) \geq 0 \\ -1 & x_i(n) < 0 \end{cases} \quad (14)$$

It is first necessary to determine the mean value of ZCR over all the frames for every epoch. Thereafter, the value is then taken over all epochs and averaged to attain the final value of the ZCR.

4.2.3. Chroma

The chroma feature is a descriptive term that condenses the tonal corpus of a musical audio signal. The terminology chroma feature or chromogram in music refers to the twelve various “pitch classes”. Chroma-based characteristics, also known as “pitch class profiles”, are useful for evaluating music with meaningfully classified pitches (frequently divided into twelve various subdivisions).

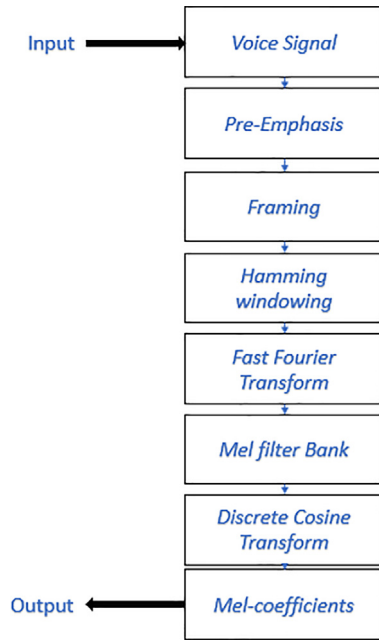


Fig. 3. MFCCs computation block diagram.

One of the most essential advantages of chroma-related features is their ability to capture harmonic and melodic components of music while being resistant to timbre and instrumentation changes. As a consequence, chroma characteristics might be considered as a requirement for high-level semantic analysis such as chord recognition or harmonic similarity estimates [61].

4.2.4. Mel spectrogram

A “spectrogram” is a pictorial illustration of the frequency content of a signal across time. The Mel scale provides a validated model for the vocal tract and is linked to Hertz by the formula below [62]:

$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right) \quad (15)$$

where m and f represent both “Mels” and “Frequency”, respectively. The Mel spectrogram is utilized to convey sound information to the models that are comparable to what a human would hear.

4.2.5. Root mean square

It represents a measurement of the volume of an acoustic stream. It is determined by computing the “square root of the sum of the mean squares of the amplitudes of the sound samples”. The RMS is calculated as illustrated in the following equation [63]:

$$x_{rms} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} \quad (16)$$

4.3. Feature selection

The main task of the feature extraction is to transfer speech emotion data from audio files into a set of features. In this work, feature extraction methods are applied on the three datasets (i.e., Emirati, RAVDESS, and SAVEE) as follows: for ‘Emirati-accented’, the MFCCs are applied based on 40-coefficient and the number of extracted features is 40. For ‘RAVDESS’, the Chroma, Mel, and the concatenation of “delta-delta, MFCCs-delta, and MFCCs”, are applied and the number of the extracted features is 260. For SAVEE

dataset, the MFCCs, Chroma, ZCR, RMS, and MEL, are applied and the number of the extracted features is 182. Feature extraction methods assigned for each experimented dataset are considered based on their frequent use for the same dataset in the literature [64–66,38,67,68]. Since there is a lack in certain features that are constantly used in the literature for building speech emotion recognition system along with the huge number of representative features for speech data, this made feature selection algorithm indisputable.

Feature selection is a pruning process to eliminate the unneeded, irrelevant, and noisy features from raw datasets such as speech emotion data. After eliminating these features, the useful and informative features are then involved in constructing the new dimension speech dataset. Feature selection is not fully investigated in speech emotion recognition when compared with other fields such as facial emotion recognition [69], text mining [70], electroencephalogram (EEG) [71], image classification [72], and cancer classification [20]. In this paper, a new feature selection method is proposed based on grey wolf optimization (GWO) algorithm in order to boost the classification performance of speech emotion recognition system. The procedure of the proposed method is illustrated in the subsequent section.

4.3.1. Feature selection method using GWO-KNN

This section provides a full elaboration of the proposed GWO methodology with k-nearest neighbors algorithm (KNN) to find the most informative and distinctive emotion features that lead to obtain more accurate speech recognition system. The proposed method mainly consists of three steps: feature extraction, feature selection, and classification. The main function of feature extraction step is to convert the speech audio files into speech emotion features. As explained before, not all features are important and significant to characterize or classify each emotion audio file. Feature selection step has been widely used and significantly contributed to pattern recognition applications. In this work, GWO is adapted as feature selection method and KNN is used for the classification task. The GWO-KNN implementation steps for emotion feature selection problem are: initialization, evaluation, updating GWO population, and stopping criterion. Two main components should be defined in advance to make GWO algorithm adaptable for feature selection optimization problem, which are solution representation and fitness function. Solution representation is explained in the initialization step, while fitness function is explained in the evaluation step. The overall methodology of the proposed method is presented in Fig. 4. The main GWO-KNN steps for solving emotion feature selection problem are thoroughly clarified below:

Step1: Initialization

This step involves the initialization of the number of iterations. Here, each wolf represents a standalone solution for a feature selection problem. Feature selection problems are categorized as combinatorial problems in which its search space is created by candidate feature subsets [73,74] in the area of optimization. By growing number of features, the feature search space is extended, therefore making it more complex. To represent it mathematically, if N stands for the number of features, there are 2^N candidate subsets of features. The problem formulation is plotted in Fig. 5.

$$\text{GWOP} = \begin{bmatrix} x_1^1 & x_2^1 & \cdots & x_D^1 \\ x_1^2 & x_2^2 & \cdots & x_D^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^n & x_2^n & \cdots & x_D^n \end{bmatrix} \quad (17)$$

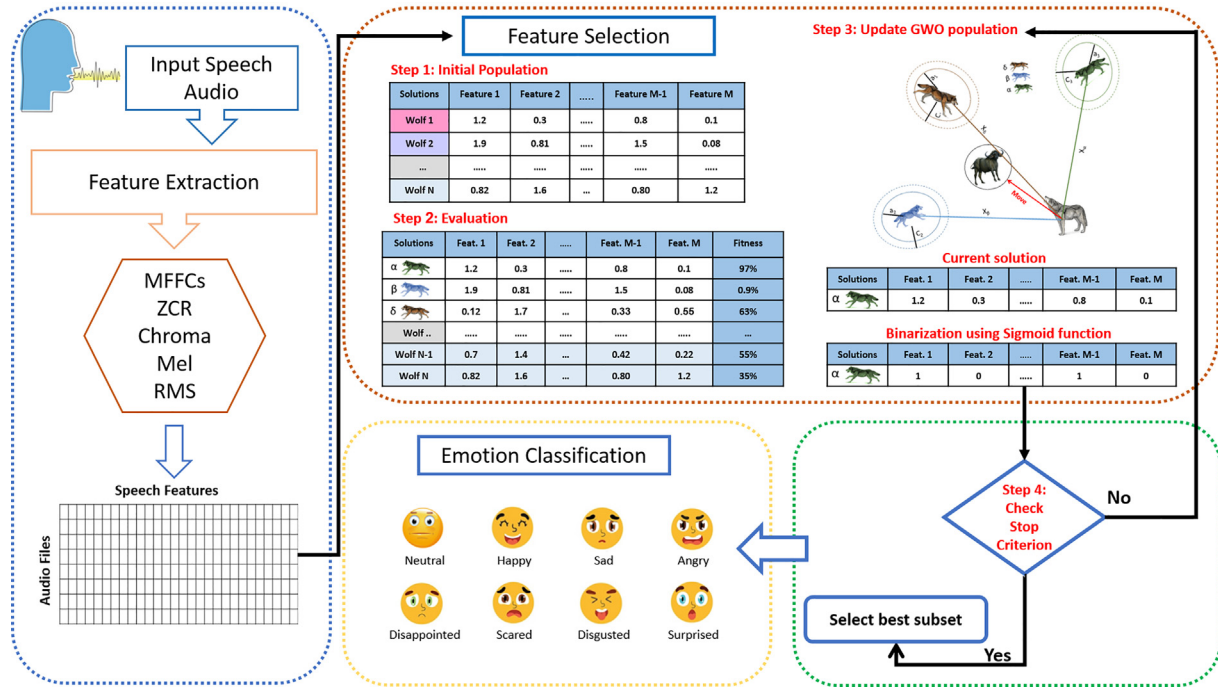


Fig. 4. Flowchart of the proposed method.

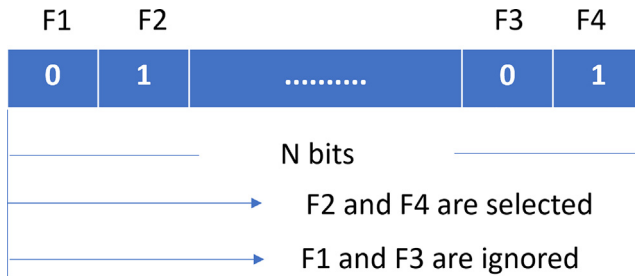


Fig. 5. The problem formulation.

subject to:

$$x \in \{0, 1\}$$

Each candidate solution (i.e., solution x), as shown in Eq. 17, introduced for feature selection problem is expressed as a binary vector of length D , $x = (x_1, x_2, \dots, x_D)$, where the feature can be maintained, assuming that the corresponding bit x , in the candidate solution equals 1, if not, the feature is discarded. In GWO, each x is called location and each bit in the solution x is called position. GWO searching procedures are launched by formulating n wolves as random binary vectors.

Step2: Evaluation

In this stage, all wolves are evaluated depending on their location vectors, where every position in the location vector is either 1 or 0. Positions that possess the value of 1 show that these positions are involved in constructing the new reduced data. After that, the reduced data is divided into training and testing in which KNN classifier is learned from the training data and evaluated in the testing data. In this research, the train-test splitting ratio is 90:10 and wolf's fitness is computed using the classification accuracy as described below.

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (18)$$

where Acc represents the objective function (classification accuracy), and TP, TN, FP, and FN represent the true positive, true negative, false positive, and false negative, respectively. The three major fitness values are X_α , X_β , and X_δ , respectively. This solution taxonomy is inspired by the social hierarchy of wolves, as explained in Algorithm 1.

Algorithm 1: Social hierarchy component of the proposed GWO

```

while ( $e < Max_e$ ) do
  for each solution ( $j$ ) do
    Calculate the fitness of the solution
     $X_\alpha$  = the fittest solution
     $X_\beta$  = the second-best solution
     $X_\delta$  = the third-best solution
  end for
   $e = e + 1$ 
end while
Return  $X_\alpha, X_\beta, X_\delta$ ;

```

Step3: Update GWO population

After the initialization of Wolves, GWO navigates the search space of feature selection problem by four main operators, including searching for prey (exploration), attacking prey (exploitation), surrounding the prey, and hunting mechanism, as illustrated in Section 3. In the original GWO, the population is updated based on Eqs. (1)–(12), as mentioned in Section 3. To achieve this purpose, the distance between the current solution and the top hierarchy solutions (i.e., X_α , X_β , and X_δ) in the population is calculated to be used in updating the location of the current solution. In practice, the process of updating each single decision variable passes through three variables which are (X_1), (X_2), and (X_3). (X_1) is generated using X_α and other variables as expressed by Equations:

[5, 3, and 4, and 8], X_2 is produced using X_β as expressed by Equations [6, 3, and 4, and 9], and finally X_3 is generated using X_δ as expressed by Equations [7, 3, 4, and 10]. At last, the decision variables of X_1 , X_2 , and X_3 are aggregated using the mean to assign new value for the decision variable in the new solution. This is an iterative process for all decision variables until constructing the new solution $X(e+1)$. It should be noted that the decision variables of $X(e+1)$ are continuous and they are transformed into binary values by using the equations [19 and 20]. The evaluation of the new solution $X(e+1)$ is similar to the evaluation procedure discussed in Step 2.

Algorithm 2: Pseudo code of the proposed GWO-KNN

Input n: No. of wolves/solutions, Max_e : No. of iterations, D_1 : Training emotion dataset, D_2 Testing emotion dataset

Output The best wolf location with best fitness

The position of n wolves is initialized randomly between 0 and 1.

```

while( $e \leq Max_e$ )      Compute the fitness of the solution
     $X_\alpha$  = the fittest solution
     $X_\beta$  = the second-best solution
     $X_\delta$  = the third-best solution
    for each solution (j)
        transfer the Wolf's to binary vector using Eq. 19 and
20
    end for
    update the control parameters of the wolves using Eq.
5–11
    evaluate the position of wolves by building a
classification model on  $D_1$  using KNN and then test it on  $D_2$ .
    update  $X_\alpha$ ,  $X_\beta$ , and  $X_\delta$ 
end while
Return  $X_\alpha$ 
  
```

$$\text{sigmoid}(X(t+1)) = \frac{1}{1 + e^{-X(e+1)}} \quad (19)$$

$$X(e+1) = \begin{cases} 1 & \text{if } \text{sigmoid}(X(e+1)) > U(0, 1), \\ 0 & \text{Otherwise} \end{cases} \quad (20)$$

Step4: Check the stop criterion

The step 3 is an iterative process that tends to achieve a better search around the best solution. This iterative process is controlled by the stopping condition (which commonly sets the maximum number of iterations). Once the stopping condition is met, the algorithm returns the best solution that carries the distinct and informative features that lead to more accurate speech emotion recognition system.

5. Data preparation

The proposed model was trained and evaluated using three different datasets: Emirati Dataset (ED), Ryerson Audio-Visual Database of Emotional Speech and Song dataset (RAVDESS), and Surrey Audio-Visual Expressed Emotion dataset (SAVEE). Each dataset was divided into training and testing datasets. The approach used in this research was text-independent and speaker-independent, where the testing dataset is different from the training dataset. To develop the models of emotion recognition, we built a model for each emotion. Each model was trained using the training dataset of the corresponding emotion. The models

then tested using the testing datasets. Regarding the ED dataset, 31 speakers were used, each uttered 8 unique sentences. There were 6 emotions, and each sentence was repeated 9 times. So, the total number of files would be 31 multiplied by 6 multiplied by 8 multiplied by 9 which gave 13,392 files. Out of the total files, 12,050 files were selected for training the models and the remaining files were used to evaluate/test the models. For the RAVDESS dataset, the total number of files was 7,356 files. However, we only used the audio files, which were 1,440 files. For training, we used 1,296 files, thereby leaving 144 files used for testing. The SAVEE dataset contains 6 different emotions in addition to the neutral emotion. Each emotion contains 60 utterances except the neutral emotion which has 120 utterances. In this research work, the training phase contains 432 utterances while the testing phase contains 48 utterances.

6. Results and discussion

In this work, all the methods are implemented using Python programming language. The core idea of this work is to propose GWO-KNN as feature selection method to optimize emotion features and thus enhance the emotion recognition performance. The performance of the proposed method is tested on three diverse emotion datasets including Emirati-accented, RAVDESS, and SAVEE. The evaluation measurements used in this study are classification accuracy, precision, recall, and F1-score. Moreover, three experiment settings are performed in this study. The first experiment involved a comparison among baseline classifiers to select the best one for the subsequent experiment. In the second experiment, a comparison has been conducted between GWO-KNN and other optimization feature selection methods. In the third experiment, GWO-KNN is compared against state-of-the-art methods that used the same datasets.

6.1. Experiment 1: effect of classifiers on emotion databases

In this experiment, some classifiers were implemented for speech emotion recognition including KNN, Multilayer perceptron (MLP), Decision Tree (DT), Logistic regression (LR), Naive Bayes (NB), and Support Vector Machine- Radial Basis Function (SVM-RBF). The input of these classifiers are the features extracted from speech data. This research uses different feature extraction methods for each emotion dataset as explained in Section 4.3. The dimension of Emirati-accented, RAVDESS, and SAVEE databases is 40, 260, and 182, respectively. Table 2 shows the results based on classification accuracy (ACC), precision, recall, and F1-score, which can be computed as, respectively, as follows:

$$ACC = \frac{TA + TR}{TA + FA + TR + FR} \times 100 \quad (21)$$

$$Precision = \frac{TA}{TA + FA} \quad (22)$$

$$Recall = \frac{TA}{TA + FR} \quad (23)$$

and

$$F1 - score = 2 \times \left(\frac{Recall \times Precision}{Recall + Precision} \right) \quad (24)$$

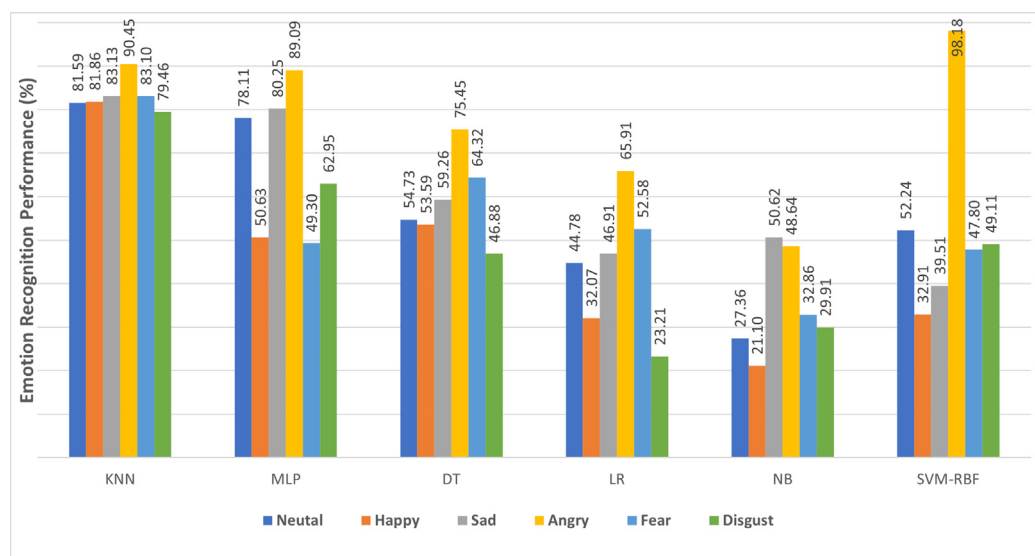
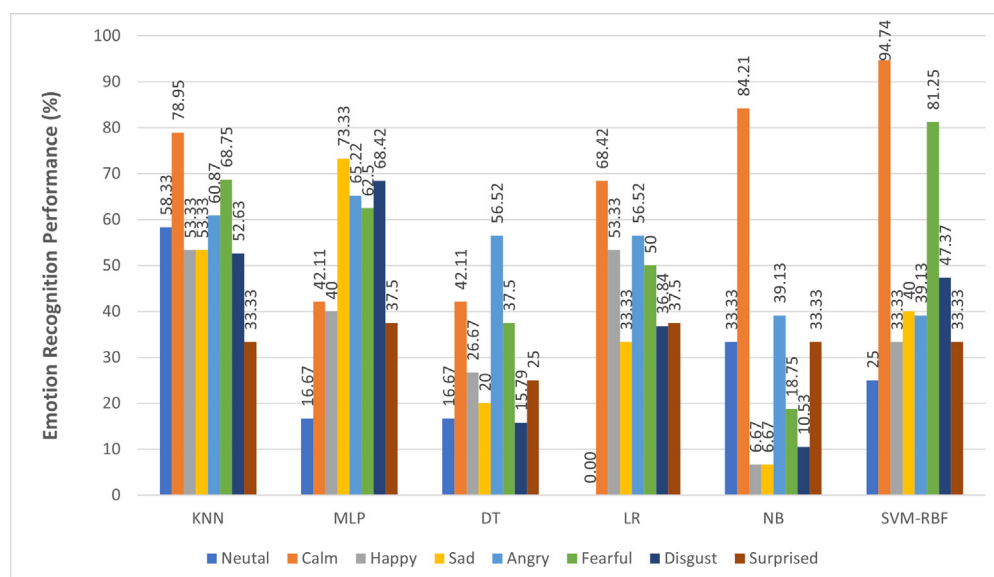
where FR , FA , TA , and TR represent the false reject, false acceptance, true acceptance, and true rejection, respectively.

Moreover, the recognition rate for each emotion using all classifiers is plotted in Figs. 6–8 for each dataset. It can be observed from Table 2 that KNN surpasses the other classifiers on two datasets (i.e., Emirati-accented and RAVDESS) on all evaluation

Table 2

Classification performance of six classifiers over Emirati-accented, RAVDESS, and SAVEE databases.

Dataset	Measure	KNN	MLP	DT	LR	NB	SVM-RBF
Emirati-accented	ACC(%)	83.25	69.58	58.37	58.37	35.27	52.69
	Precision(%)	83.16	71.35	58.39	58.39	36.04	81.80
	Recall(%)	83.26	70.27	58.41	58.41	35.08	53.13
	F1score(%)	83.19	69.29	58.30	58.30	34.89	56.50
RAVDESS	ACC(%)	56.64	48.25	31.46	44.05	30.76	49.65
	Precision(%)	56.59	51.91	29.76	39.62	31.72	52.70
	Recall(%)	57.44	48.11	30.41	41.99	29.07	49.26
	F1score(%)	55.99	44.40	29.87	39.92	26.18	47.23
SAVEE	ACC(%)	65.95	72.34	53.19	65.95	51.06	27.65
	Precision(%)	63.29	69.01	47.61	61.49	54.54	04.12
	Recall(%)	57.99	64.20	48.55	61.90	42.85	13.26
	F1score(%)	58.72	64.95	46.14	60.82	42.01	06.29

**Fig. 6.** Emotion recognition performance assessment using Emirati-accented database based on KNN, MLP, DT, LR, NB, and SVM-RBF.**Fig. 7.** Emotion recognition performance assessment using RAVDESS database based on KNN, MLP, DT, LR, NB, and SVM-RBF.

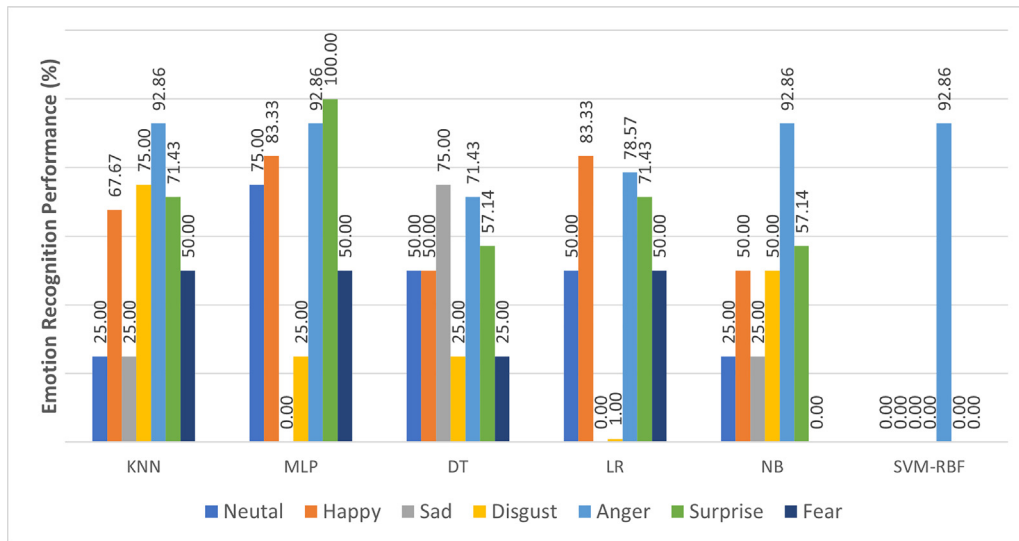


Fig. 8. Emotion recognition performance assessment using SAVEE database based on KNN, MLP, DT, LR, NB, and SVM-RBF.

measurements. In Emirati-accented, the difference between KNN and the second best (i.e., MLP) is 13.67%. In RAVDESS, the difference between KNN and the second best (i.e., SVM-RBF) is 6.99%. On the other hand, for SAVEE dataset, MLP achieves the best result with 6.39% difference over KNN. Fig. 6 demonstrates the recognition performance of all classifiers for all emotions on Emirati-accented database. kNN yields accuracy greater than 79% for all of the emotions and also attains the highest recognition rate for all emotions except Angry. The best results for recognizing Angry emotion are obtained by SVM-RBF.

Recognition rate yields for various emotions for all the classifiers with RAVDESS is shown in Fig. 7. kNN and MLP achieve the best recognition performance. However, kNN has a better recognition stability than MLP since the latter shows a substantial deteri-

oration in recognition performance for neutral, calm, and happy. The lowest classification rate obtained by kNN is 33.33% for surprised emotion, while MLP yields 16.67% for neutral emotion. The recognition performance of SAVEE dataset for seven emotions is illustrated in Fig. 8. It can be observed that MLP achieves better or similar recognition compared to other classifiers for neutral, happy, disgust, anger, surprise, and fear. The lowest recognition result is for sad. Overall, the best recognition performance is attained by kNN; therefore, it will be chosen for the subsequent experiment that is mainly for improving emotion recognition system.

6.2. Experiment 2: comparison between GWO-KNN and other approaches

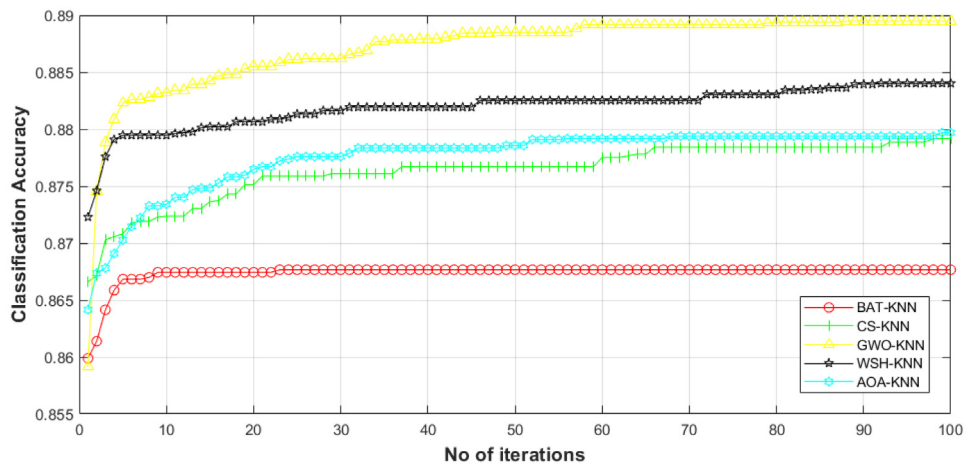
In this experiment, the performance of the proposed method has been compared against well-known and recent optimization algorithms, which are Bat Algorithm (BA) [75], Cuckoo Search (CS) [76], White Shark Optimizer (WSH) [77], and Arithmetic Optimization Algorithm (AOA) [78]. It should be noted that the compared algorithms are adapted as feature selection methods, and since these algorithms are stochastic in nature, they are performed 10 times and the results are calculated using the mean to avoid bias in the results. The parameter settings of the above

Table 3
Parameter settings of optimization algorithms.

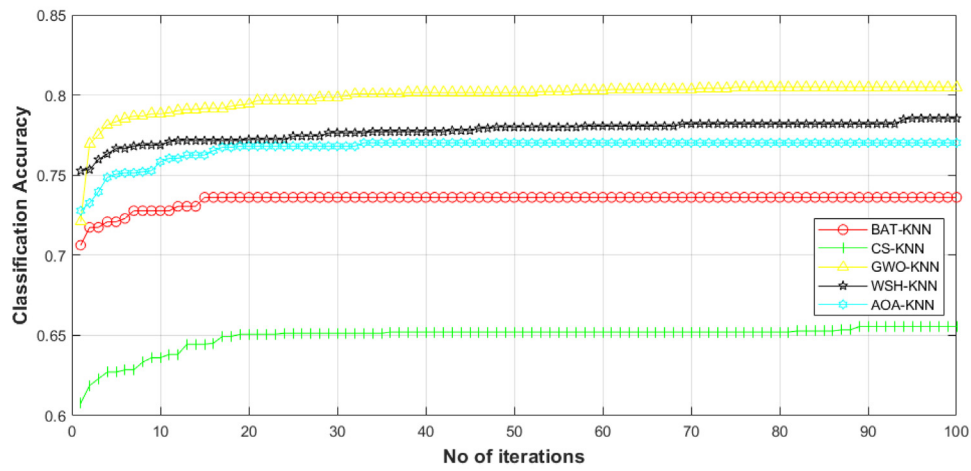
Approach	Parameters
CS	$\beta=1.5, P_a=0.25$
BAT	$A=0.5, r=0.5$
AOA	$\alpha=5, \mu=0.5$
WSO	$f_{min}=0.07, f_{max}=0.75, a_0=6.25, a_1=100, a_2=0.0005, \tau=4.125$
GWO	-

Table 4
Results of comparison between GWO-KNN and other optimization feature selection methods.

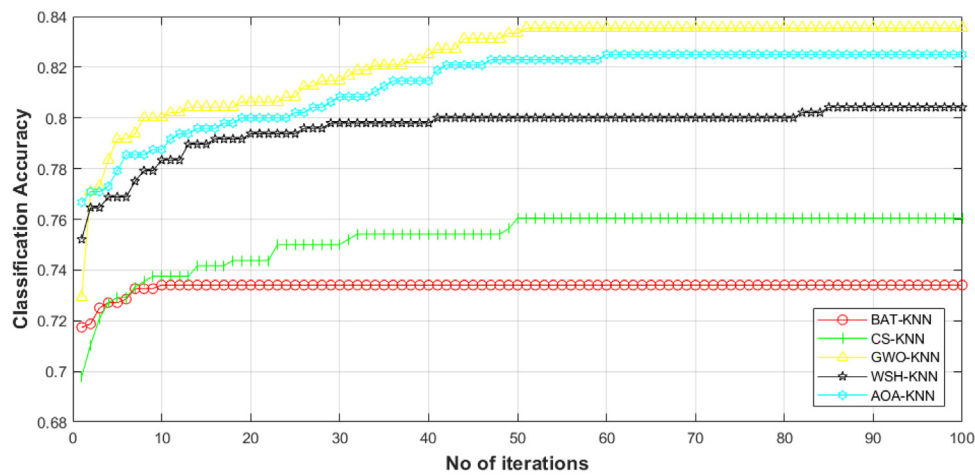
Datasets	Measurments	KNN	GWO-KNN	BAT-KNN	CS-KNN	WSH-KNN	AOA-KNN
EM	Acc	83.25	88.95	86.77	87.92	88.40	87.97
	Precision	83.16	88.93	86.72	87.88	88.38	87.92
	Recall	83.26	88.96	86.80	87.91	88.42	87.95
	F1score	83.19	88.93	86.73	87.87	88.38	87.92
	T-Sig.			*	*	*	*
RAVDESS	Acc	56.64	80.48	73.61	65.55	78.54	77.01
	Precision	56.59	79.74	73.29	66.45	77.85	76.99
	Recall	57.44	80.75	73.99	65.70	78.62	77.17
	F1score	55.99	79.93	73.09	64.66	77.84	76.49
	T-Sig.			*	*	*	*
SAVEE	Acc	65.95	83.54	73.40	76.04	80.41	82.50
	Precision	63.29	83.35	72.85	75.37	81.12	82.56
	Recall	57.99	80.16	73.53	73.40	76.48	78.71
	F1score	58.72	80.45	72.70	72.53	76.73	78.65
	T-Sig.			*	*	*	*



a) Convergence rate of Emirati dataset

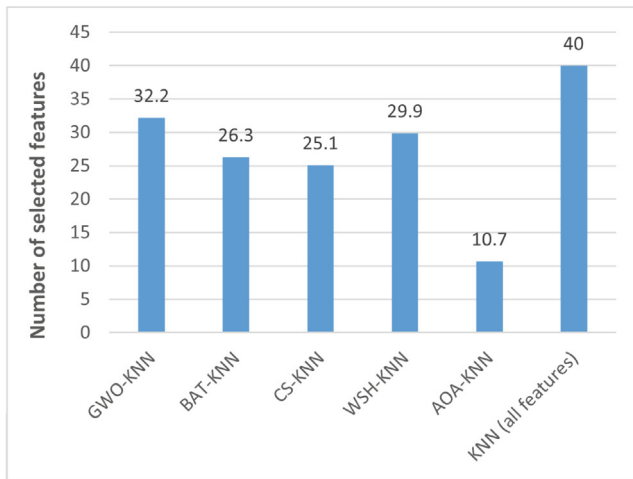


b) Convergence rate of RAVDESS dataset

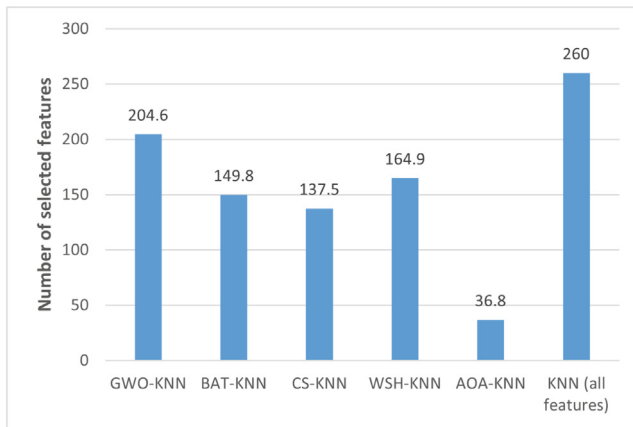


c) Convergence rate of SAVEE dataset

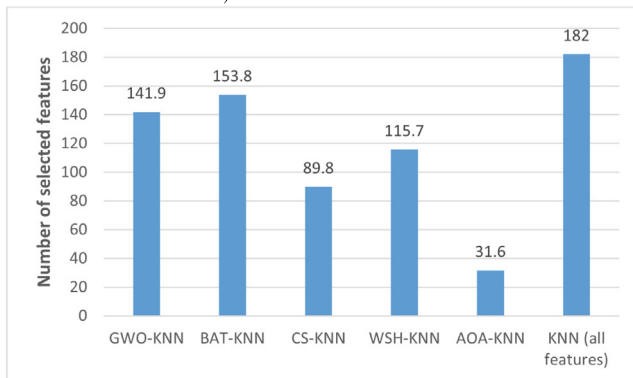
Fig. 9. Convergence rate of GWO-KNN and other optimization algorithms.



a) Emirati dataset



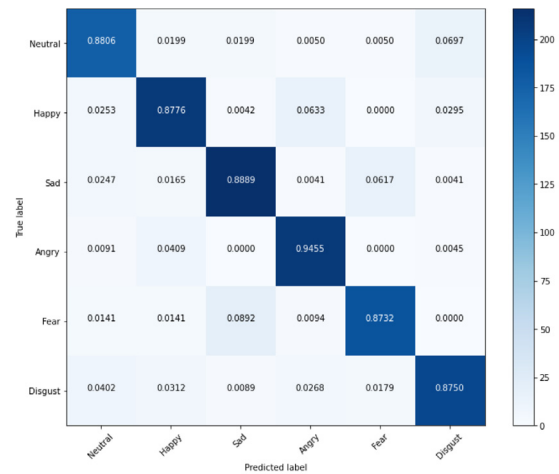
b) RAVDESS dataset



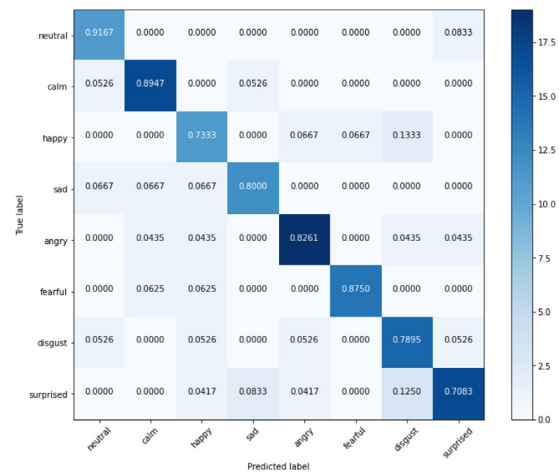
c) SAVEE dataset

Fig. 10. The number of features selected by GWO-KNN and other optimization algorithms.

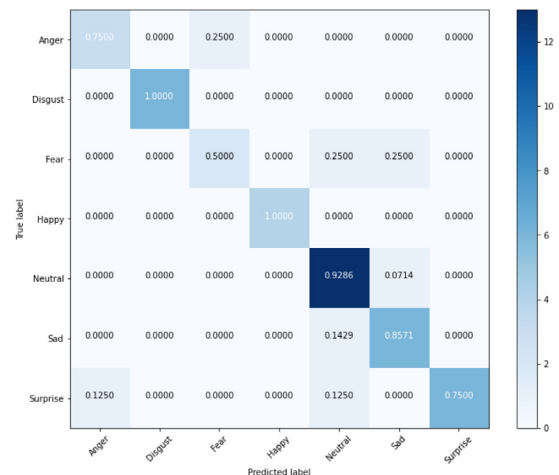
optimization algorithms are presented in Table 3. The performance of GWO-KNN and other optimization feature selection methods (i.e., BAT-KNN, CS-KNN, WSH-KNN, and AOA-KNN) are evaluated using the same measurements that were used in the previous experiment, as presented in Table 4. In addition, the results between GWO-KNN and other methods are statistically tested using Wilcoxon signed-rank statistical test [79], and it is symbolized as $T - Sig$ in the same Table. With the probability value at $\alpha \leq 0.05$, “*” indicate that the results yielded by GWO-KNN are statistically significant, while “-” indicates that there is no statistical significant results in favour of GWO-KNN.



a) Emirati dataset



b) RAVDESS dataset



c) SAVEE dataset

Fig. 11. Confusion matrices of GWO-KNN for the three different datasets.

From Table 4, it has been observed that GWO-KNN excelled all other feature selection methods on all the evaluation measurements for all datasets. In addition, the results of Wilcoxon signed-rank statistical test, demonstrate that GWO-KNN achieves

statistically significant results when compared with other methods. The convergence behaviour of GWO-KNN and other methods is plotted for all experimented emotional data, as plotted in Fig. 9. It can be inferred that the convergence trend of GWO-KNN is significantly better than all methods. The number of the selected features by all optimization feature selection methods is presented in Fig. 10, it can be observed that AOA-KNN produces the lowest number of features in all datasets; however, they are not associated with the highest classification accuracy. It can be noted that reducing the size of the feature subset to a minimal size leads to a downgrade in the classification accuracy. The confusion matrices attained by KNN classifier for three different datasets are shown in Fig. 11.

6.3. Experiment 3: comparison with other state-of-the-art approaches

In this experiment, the performance of the proposed method is assessed against other state-of-the-art methods in the literature, as shown in Tables 5–7. It should be noted that the classification accuracy of GWO-KNN is reported based on the best results from 10 runs, since the results of state-of-the-art methods are based on the same criteria. Table 5 reports the performance based on classification accuracy for GWO-KNN and other methods using Emirati speech database. Most of the methods used MFCCs to extract the features from speech audio files but with different classifiers. It can be concluded that GWO-KNN surpasses all other methods in classifying the six emotion states of the Emirati speech data with high classification accuracy (89.17%). Table 6 reports the performance of GWO-KNN and other methods using RAVDESS. Different feature extraction methods and classifiers were applied on RAVDESS dataset. The best emotion classification result (81.25%) is achieved by GWO-KNN. The results of classifying emotion states of SAVEE dataset using GWO-KNN and other emotion classification methods are presented in Table 7. The feature extraction method and the classifier used by each method are also presented in the same table. It is clear that GWO-KNN is superior to other methods with high classification accuracy (85.42%). These satisfactory results are owing to the powerful searching operators of GWO

Table 5
Comparison of the proposed method based on the recognition performance on the Emirati speech database.

Reference	Classifier used	Features used	Obtained accuracy
Shahin et al. [21]	GMM-DNN	MFCCs	83.97%
Shahin [80]	CSPHMM3s	MFCCs	77.80%
Nassif et al. [81]	CNN	MFCCs	78.52%
Proposed method	KNN	MFCCs	89.17%

Table 6
Comparison of the proposed method based on the recognition performance on RAVDESS corpus.

Reference	Classifier used	Features used	Obtained accuracy
Bhavan et al. [32]	Bagged ensemble of SVMs	MFCCs, spectral centroids and MFCC derivatives	75.69%
Zeng et al. [82]	DNNs	Spectrograms	64.52%
Shegokar and Sircar [83]	SVMs	Continuous wavelet transform, prosodic coefficients	60.1%
Shahin et al. [38]	CNN	MFCC derivatives	75.3%
Kwon [84]	Deep stride CNN	High level features	81.00%
Proposed method	KNN	MFCC derivatives, Chroma, and Mel	81.25%

Table 7

Comparison of the proposed method based on the recognition performance on SAVEE corpus.

Reference	Classifier used	Features used	Obtained accuracy
Mao et al. [85]	CNN	Automatically learned by CNN	73.6%
Zhang [9]	Logistic regression	OpenSMILE-2.3.0 (waveform, Mel/Bark spectrum, FFT spectrum, etc.)	60.16%
Özseven [25]	SVM	OpenSmile features	72.39%
Liu et al. [86]		MFCC features	76.40%
Hajarolasvadi and Demirel [87]	3D CNN-Based approach with K-means Clustering	Deep feature of Neural Network	81.05%
Proposed method	KNN	MFCC, Chroma, ZCR, RMS, and Mel	85.42%

guided by searching for prey (exploration), encircling prey, attacking prey (exploitation), and hunting mechanism operators that effectively steer the optimization process to the most promising speech emotional subset of features that carry the most interesting/useful patterns for emotion states classification.

7. Conclusion and future work

Building an emotion recognition system using speech signal data is an active research area in human computer interaction, which has been used to analyze and classify the emotional states of human speech content. In this paper, we proposed an automatic speech emotion recognition based on GWO as a feature selection method and KNN for emotion classification. Initially, feature extraction is applied on speech signal data and then is passed to GWO to find the most relevant emotion features and discard the irrelevant and redundant features. The reconstruction of the input speech data with such relevant and meaningful acoustic features lead to an accurate and robust emotion recognition system. This is owing to the capability of GWO in efficiently exploring the emotional feature space and finding the best features that carry rich patterns for emotion classification. The performance of the proposed method is tested using the Emirati-accented, RAVDESS, and SAVEE databases. The evaluation measurements used in this study are classification accuracy, precision, recall, and F1-score. To validate the performance of the proposed method, four optimization feature selection methods were implemented (i.e., BAT-KNN, CS-KNN, WSH-KNN, and AOA-KNN), and their results were compared with GWO-KNN. The results demonstrated the superiority of GWO-KNN over other comparable methods. In addition, GWO-KNN was compared against the state-of-the-art methods using the same datasets. The experimental results showed that GWO-KNN provided better results over current state-of-the-art emotion recognizes. As GWO-KNN provides very promising outcomes for emotion recognition system, in the future, it can be further investigated from various perspectives:

1. Enhancing the search capability of GWO through hybridization with other local search-based algorithms or adding efficient and novel searching operators.
2. Applications with different types of speech pathologies such as dysarthria, expressive disorders, articulation disorders, and fluency disorders.

CRedit authorship contribution statement

Ismail Shahin: Writing – original draft, Data curation, Validation, Formal analysis, Visualization. **Osama Ahmad Alomari:**

Conceptualization, Methodology, Software, Writing – original draft. **Ali Bou Nassif**: Writing – original draft, Data curation, Validation, Formal analysis, Visualization. **Imad Afyouni**: Writing – original draft, Data curation, Validation, Formal analysis, Visualization. **Ibrahim Abaker Hashem**: Writing – original draft, Data curation, Validation, Formal analysis, Visualization. **Ashraf Elnagar**: Writing – original draft, Data curation, Validation, Formal analysis, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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